

The Universal Mobile Telecommunications System — a mobile network for the 21st century

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The requirements which the third generation mobile network will be required to deliver have been identified by Groves and Clapton [1]. In considering how these requirements can be met, firstly the types of networks (both fixed and mobile) that will be available in that time frame have to be considered. On the fixed network side, there is expected to be widespread use of intelligent networking, with fully digital networks supporting broadband and narrowband ISDN access as well as POTS. If the Universal Mobile Telecommunications System (UMTS) is to be commercially successful then it will be necessary for both mobile and fixed network operators to identify the benefits it gives and its role in meeting customer needs. It is also important for the operators to ensure that they are getting sufficient financial return on the massive investments made in GSM and the fixed networks. All of these considerations therefore come into play in considering what UMTS will be and how it will be provided. This paper will concentrate mainly on the technical issues surrounding UMTS, but will touch upon commercial aspects in discussing how operators could migrate or evolve from their current networks to a future UMTS network.

1. Introduction

The Universal Mobile Telecommunications System (UMTS) is currently being developed within Europe as a candidate third generation system for mobile telecommunications. It will complement and, in many cases, provide integration with the Global System for Mobile communications (GSM), which is currently in service within Europe and increasingly world-wide. In order for UMTS to be commercially viable it must:

- provide broadband flexible bandwidth allocation, supporting multimedia and Internet services at rates of up to 2 Mbit/s, and supplementary services independently of whether the customer is fixed or mobile in a manner that appears transparent to the customer,
- integrate the provision of cordless, paging and broadband mobile services into one system — UMTS will be provided in business, domestic, vehicular and (multi-operator) public environments,
- support mass-market service provision efficiently and at a low cost,
- be a global system allowing global roaming and global service provision (current systems tend to be regionally biased, although GSM has enjoyed an increasing level of success globally).

The resultant UMTS system will provide network and service infrastructure convergence for fixed and mobile networks by allowing network operators to offer service to

customers flexibly and efficiently, irrespective of their means of access. This does not imply that fixed and mobile networks will cease to exist in their own right when UMTS is introduced, rather that UMTS will supplement existing systems such as GSM.

This paper introduces general cellular concepts and efficiency improvements that could be made over current second generation systems, the proposed high level UMTS network architecture and the different evolution paths to UMTS.

The paper is based upon ongoing UMTS standards work in the European Telecommunications Standards Institute (ETSI) and the results of the RACE Mobile Networks (MoNet) project [2].

2. What makes cellular networks different?

In order to understand any design for a cellular network it is important to understand what makes cellular networks different from conventional fixed networks. Cellular networks differ from fixed networks in two main respects — the first and most obvious differentiator is that each customer may move location; the second discriminator is that there is a scarcity of radio spectrum available for use by a network operator to communicate with its customers.

It is the limited radio spectrum that gives rise to the partitioning of the network into small cells, in order that the subset of the total bandwidth allocated to an individual cell

can be reused repeatedly throughout the cellular network. To support the 'blanket' network of cells, each with a transceiver, a large transmission network infrastructure is required consisting of radio resource controlling nodes and switching nodes. In GSM these nodes are called the base station transceiver (BTS), base station controller (BSC) and mobile switching centre (MSC) respectively.

As the customer may move freely around cells both within his home network, and in the case of GSM in other visited networks, there is the need to ensure that service is provided irrespective of where the customer is and in a manner that is transparent to the customer. Conventional fixed networks use the fixed point of attachment at the switch to determine the services both subscribed to by the customer and for delivery of calls. As this approach is obviously not appropriate for the support of cellular customers, a number of generic features, described at a high level below, exist within cellular networks in order to offer secure and transparent service for subscribers on the move. The generic descriptions are based upon the UMTS functional architecture, but are valid for most cellular networks including GSM.

- **Location management/location updating** — location management is the generic term applied to procedures that maintain location information within a cellular network. The cellular network is divided into administrative domains called location areas. The location area identifies the approximate geographical area, including the country and the network where the customer is located, in order to simplify call delivery. When a customer changes location area, the mobile terminal must inform the cellular network, using the location update procedure, that the location area has been changed as no permanent connection is maintained with the network. The location area of the subscriber is stored locally to that area in a visited database along with details of the customer's subscription. When a change of location area is linked with a change of visited database, in addition to updating the location area in the new visited database, the home database is updated so that it correctly identifies the new visited database for the call delivery procedure. At the same time, the home database will transfer subscription records for the customer to the visited database in order that the subscription records are held locally, eliminating the need for long distance signalling and any potential time delay when services are activated. UMTS will extend the basic concept further by supporting service domains, in addition to location areas. The inclusion of the service domain in the location management procedures will allow the system to identify different access network service and administrative capabilities for a particular location area. This would be primarily of use to private networks interconnected to the public networks,

enabling them to pass back information about localised service support to the customer's home database for use during the call-delivery procedure. However, the feature could also be used by public network operators during the roll-out of new features into the access networks.

- **Paging, call delivery and terminating services** — when a call is delivered to a UMTS network node, the node invokes mobility processing in the intelligence layer. The mobility layer interrogates the home database in order to obtain service information concerning the subscribed customer features for terminating calls and the location area. The location area is used to derive the network routeing to the visited switch where the subscriber is most likely to be located. Terminating services are then executed within the intelligence layer (e.g. unconditional forwarding to a voice messaging platform) followed by the call being routed to the visited switch. When the call arrives at the visited switch, processing is passed to the mobility processing layer again in order to execute the final stage of call delivery — paging. Although the network knows the approximate position of the customer, i.e. the location area, the network does not know in which cell the customer is located. In order to locate the mobile, the network starts the paging procedure whereby the identity of the customer is broadcast on all cells within a location area. If the mobile terminal, where the customer is located, hears the identity of an associated customer being broadcast, it replies to the network mobility processing layer. The call is then set up to the terminal.
- **Mobile call origination and originating services** — when the mobile customer makes an outgoing call, the call processing is passed up to the mobility and service processing layer in order to check the subscription and to apply originating services. These originating services may be either standardised, offered by all networks, or network specific, in which case service processing is passed back to the home network.
- **Handover** — the handover procedure is designed to maintain both the signalling connection(s) and traffic connection (s) (if present) between the mobile terminal and the network when the subscriber moves from cell to cell. This process requires the changing of the radio frequencies used across the air interface and, if necessary, the re-routeing of connections within the fixed networks. This re-routeing of fixed links may range from a change of link between transceivers to a change of switch and even, as currently proposed for UMTS, a change of network.
- **Authentication and ciphering** — the authentication and ciphering procedures are designed in order to protect the services offered to the subscriber. The

authentication procedure takes place between the network and the UMTS identity module (UIM) associated to a mobile terminal. The network interrogates the UIM to ascertain that the subscription is not fraudulent and when satisfied provides service to the customer. It has also been proposed that, for UMTS, the UIM will be able additionally to interrogate the network to ensure that the network with which it is in communication is not fraudulent. This feature is proposed to counter a perceived threat offered against customers by private network domains incorporated in the system. The ciphering procedure encrypts data across the air interface to ensure that eavesdropping cannot occur as in older analogue systems. Both the authentication and ciphering keys are transferred from the home database to the visited database during location management procedures.

In the GSM system, separation of the switching layer from the mobility and service layer at the visited switch does not exist. As a result, all mobility processing and most service processing occurs within the visited MSC and takes approximately 50% of the processing capacity of the switch. This causes the switch to be processor limited, reducing the number of customers that can be supported in a busy hour relative to an equivalent switch supporting fixed customers. A key technical aim of UMTS is to reduce this mobility impact on the switching layer by providing separation of the switching layer from the mobility and service layer.

3. UMTS - a fixed/mobile convergent technology

One of the prime technical aims of UMTS is to separate out the mobility and service support layer from the transport and switching layer in order to allow:

- common call processing for both fixed and mobile access customers provided by the switching layer — while ATM-based switches are the target switching

layer for UMTS, since they support flexible bandwidth allocation, the functional design does not preclude the use of other switch types,

- common service processing for both fixed and mobile access customers provided by the mobility and switching layer — this will allow the same supplementary services to be offered to both fixed access and mobile access customers, with this service support also being able to cover the provision of supplementary services to GSM customers (as shown in section 6),
- mobility support for all UMTS mobile access customers, including roaming cordless terminals, in a generic manner,
- a drastic reduction in the switch processing load attributable to mobility support and an increase in switch utilisation by shifting mobility processing to service and mobility layer platforms,
- service interaction between mobility processes, such as handover, and customer services, such as Call Completion Busy Subscriber, to be minimised and simplified.

Many of the design requirements described above could have been met by the pre-phase 1 GSM proposals, shown in Fig 1; this design provided separation between the switching layer and the service and mobility layer within the core part of the network in a manner similar to the 'intelligent network' conceptual model which was developed later. The 'intelligent network' (IN) concept separates switching, performed solely by the switch, from 'intelligent' computing, performed by service control functions. Unfortunately, as the switching technology of the late 1980s/early 1990s was not able to support the VMSC-VLR interface (visited switch-mobility and service layer

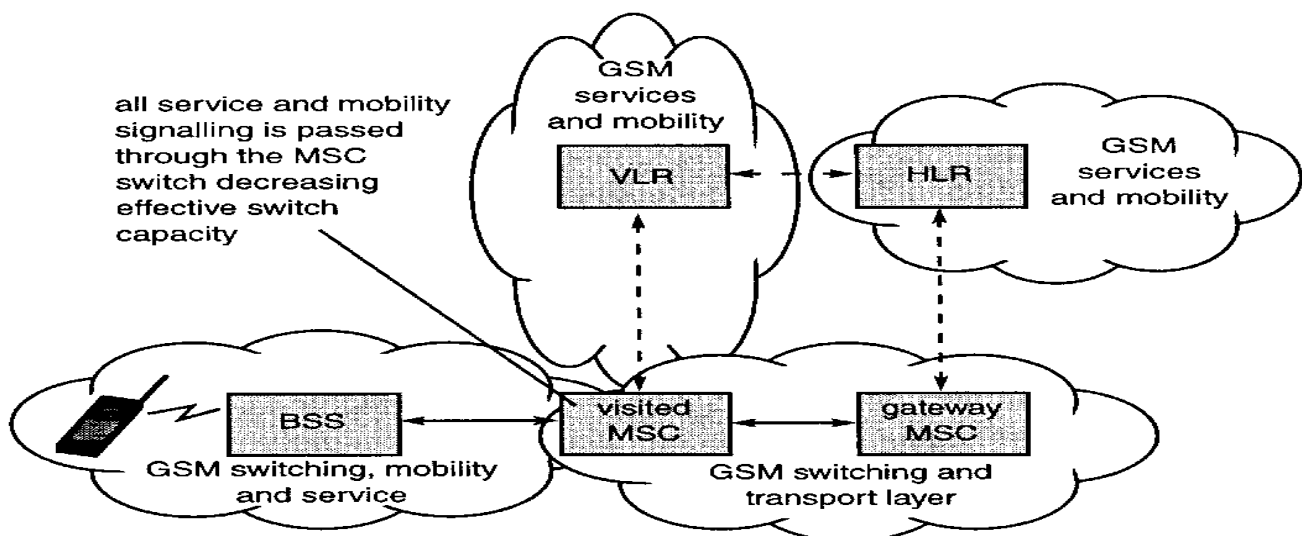


Fig 1 Pre-phase 1 GSM — separation of mobility and service layer from switching layer.

interface), the two entities were merged for implementation of the GSM system, losing the benefits described previously.

The UMTS network architecture builds upon the concepts of pre-phase 1 GSM by extending the separation between the switching layer and the service and mobility layer across the whole system. To achieve this aim, the UMTS design will develop and extend the IN concept. It will additionally recognise that the access network structure and capabilities will vary depending upon the environment that it is being served (e.g. domestic, business, satellite). The UMTS design supports the different access networks by providing a common interface structure, independent of access network type, both across the air interface and between the access networks and the core network elements as shown in Fig 2. The general structure of both interfaces will separate transport-related signalling from mobility and service signalling. The separated functional interface between the core and access networks will help minimise the functional and processing impact on the switches of mobility procedures and will allow common transport interfaces to be used at the switch. This functional separation will be mirrored within the access network and across the air interface in order to help reduce the complexity of the access network design and to allow the transport capabilities supported by the mobile terminal to appear the same to applications as those offered by fixed terminals.

In order for the UMTS architecture to become reality, the concepts it embodies need to be standardised. The architecture development will be split into four main areas:

- the definition of the overall UMTS system design,
- the development of IN to incorporate UMTS mobility support and GSM customised supplementary service support,
- the development of transport and switching standards that are suitable for the UMTS support of both fixed and mobile access,
- the definition of UMTS access networks.

4. Mobility procedures

The functional architecture provides a framework on which the mobility and service procedures are supported. The mobility procedures, however, define the type of functionality required in each of the network nodes. Of all the mobility procedures supported by mobile systems, handover has the most complex requirements, so the following section presents, with particular emphasis on the switching and intelligence requirements, how handover could be implemented for UMTS. Before describing how the procedure could be implemented, it is necessary to understand the basic handover design requirements:

- the mobile terminal can change its point of attachment to the network without user action and in a manner that appears seamless to the user,
- the load that handover applies to the cellular network should be minimised while still meeting the above requirements,

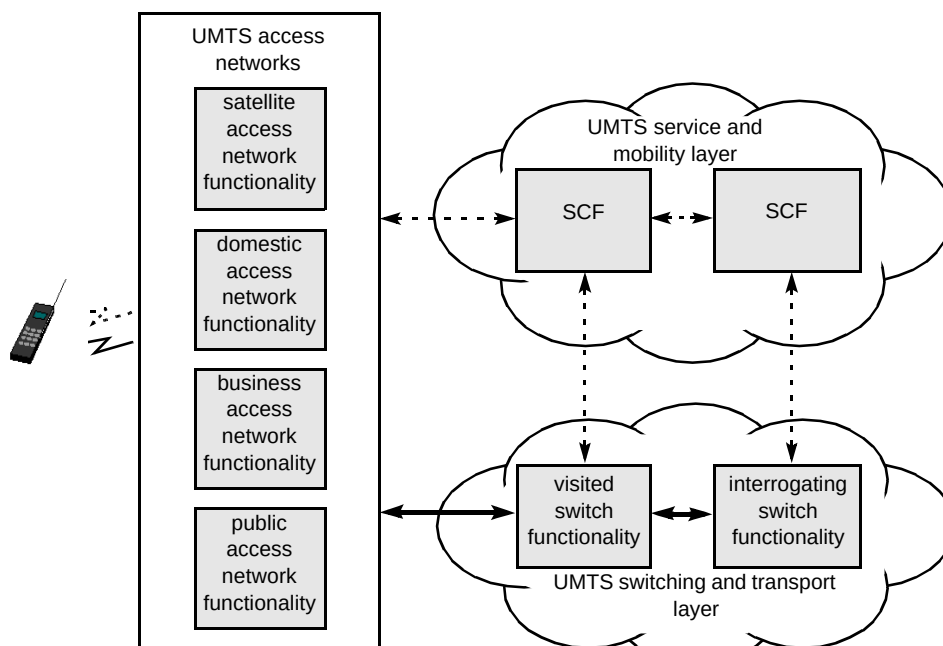


Fig 2 The UMTS functional architecture.

- it should be possible to extend the basic handover to support handover between switches both within the same network and between networks while minimising the data that needs to be held at any individual network node.

Handover is dependent upon the physical layout of the network. Therefore in the following examples two physical implementations have been assumed based upon the UMTS architecture presented. When handover occurs between cells belonging to different access network parts attached to the same switch but by different links (access networks 1 and 2 in Fig 3), it is necessary to switch both the traffic and signalling between access networks at the switch. At the same time, the impact of the mobility procedure on the

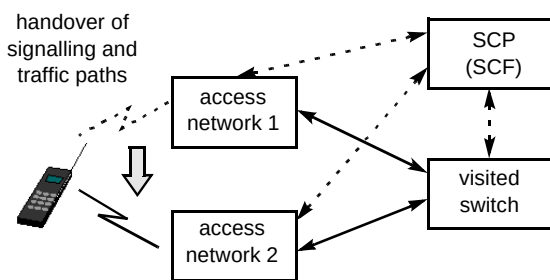


Fig 3 Handover at the switch for one possible physical configuration.

switch should be minimised, with the switch ideally only being involved when the traffic and signalling paths are physically changed. As a result a UMTS handover procedure has been designed [3] as follows:

- when an access network part identifies that a handover between cells does not take place solely within its control domain (e.g. access network 1), it passes control to the SCP to continue handover processing,
- the SCP then attempts to identify in which access network the target cell is, based upon information provided by the mobile terminal, and then ascertains whether the target cell is within its control domain — if the access network is within its control domain, it attempts to set up fixed links to the access network and requests the access network to allocate radio resources,
- the SCP then tells the mobile, through the old access network (e.g. access network 1), the radio information that it will need to re-attach on the new access network (e.g. access network 2),
- the SCP instructs the switch to bridge traffic and signalling paths passing through both the new and old access networks — shortly after, the mobile terminal re-attaches to the target cell on the new access network (e.g. access network 2),

- if the handover is successful, the SCP proceeds to clear down fixed links to the old access network and instructs the old access network to clear down radio resources which are no longer required.

If, at the second step, the SCP identifies that the target access network is attached to another switch, the SCP negotiates with other SCPs, as shown in Fig 4, in order to arrange for the allocation of the fixed and radio resources remotely. This mechanism is required as each SCP supports total control and network knowledge only within its local domain. The reasoning behind this approach is twofold — firstly it makes system data builds less complex, and secondly allows network operators to maintain control over resource allocation within their domains when handover occurs between networks.

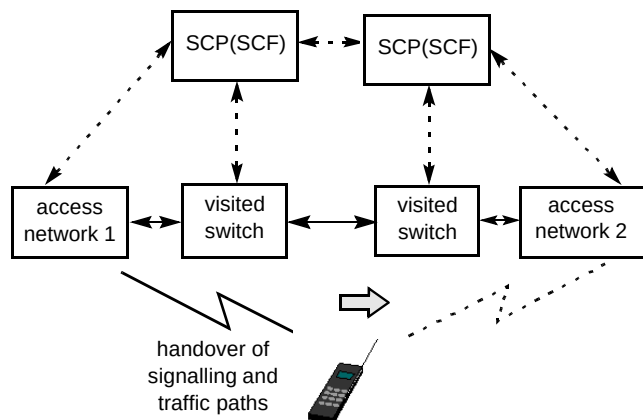


Fig 4 Handover between access networks on different switches for a particular network configuration.

By utilising the procedures described above, the load on the switch is minimised as it does not have to interpret mobility signalling. The only requirement on the switch is that the routing of a call can be changed at the switch without changing the state of the call; this will need to be accommodated in future service releases of IN and B-ISDN. The design does, however, require that there exists an intelligence interface from the access network to the SCP in order to support the mobility procedures.

5. Introduction of UMTS

Before considering how current systems can be evolved into UMTS, it is first important to consider how the UMTS system will be introduced and its relationship to existing systems. In addition, the method by which UMTS is introduced will primarily depend upon whether the access environment is public or private.

5.1 Public access

By 2005, when UMTS equipment will start becoming available, GSM/DCS1800 will be a mature system with a high market penetration, estimated to be as much as 50% of

the UK population, and will provide nearly total geographical coverage with support for voice and fixed low-bandwidth data telephony. In contrast, UMTS will support both voice and variable medium bandwidth data telephony. The support of higher data bandwidths coupled with limited radio spectrum will demand a cellular network of even greater density than those provided for GSM/DCS1800. As providing a nationwide network immediately for UMTS public access would be uneconomic, UMTS public access would be introduced in a phased approach as shown in Fig 5. The service would be rolled out to areas of highest demand first, corresponding to large densely populated urban conurbations such as London, followed by a gradual roll-out to areas of lower demand, such as rural areas like East Anglia. In areas where UMTS public access did not exist, customers with a UMTS subscription could revert to GSM public access by using dual mode GSM/UMTS handsets.

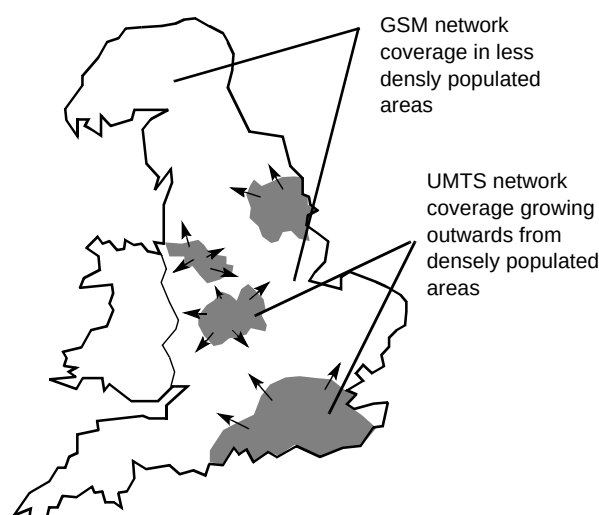


Fig 5 Roll-out of UMTS.

5.2 Private access

In private environments, e.g. domestic and business environments, UMTS access will be rolled out at the discretion of private network operators, subject to their obtaining radio licences or licence waivers from the relevant government agencies, much as with present cordless PBX systems. The private network will then obtain access to the UMTS core network at both the switching layer, as is provided today, and at the service and mobility layer, where a functional interface and a collection of UMTS support services will be provided. In this approach, the introduction of private UMTS will be the responsibility of private individuals and companies who will upgrade in order to meet their personal or business needs. As shown in the next section, UMTS private access may well be an evolution of existing private access schemes.

6. Evolution to UMTS

Evolution to UMTS can be separated into evolution from both fixed systems and cellular systems. UMTS will represent the convergence of the two evolution paths and the convergence of standards for both systems. This section will examine evolution from both types of system and highlight the barriers to evolution.

6.1 Evolution from fixed networks to UMTS

Many of the current developments within fixed network design revolve around the use of IN concepts which separate switching, performed solely by the switch, from 'intelligent' computing, performed by the service control functions (SCFs). The design of UMTS aims to extend this concept to allow mobility to be supported from an 'intelligent' computing layer of SCFs and to allow services to be executed between networks. A first step towards providing mobility from IN has been made by the developing European cordless terminal mobility (CTM) standard. This network feature, shown in the first part of Fig 6, is being developed within a EURESCOM project and standardised via ETSI. It will provide cordless handset users with the ability to roam and receive services between different wireless PBXs in a manner similar to a cellular network. In order to achieve the network mobility support required for CTM, the next phase of IN (Capability Set 2 (CS2)), will, in addition to new service features, support:

- a basic form of interconnect between access equipment and the SCFs in the core network in order to pass non-call related information between the access network and the core network,
- interconnect between SCFs of different networks allowing inter-network roaming of customers.

The functionality provided by IN CS2, while being sufficient for the first phase of CTM, which has basic mobility requirements, will require enhancement for UMTS in order to provide:

- a functionally direct means of interconnect between the access network and the SCFs in order to minimise switch loading,
- support for the full range of UMTS procedures and the support of service features described within the GSM 'CAMEL' feature [4],
- the ability to manipulate call bearers without changing the state of the call in order to support the handover procedure.

As UMTS is a broadband system, its introduction will also require broadband switches to support IN and allow switching of call legs without any impact on the call in

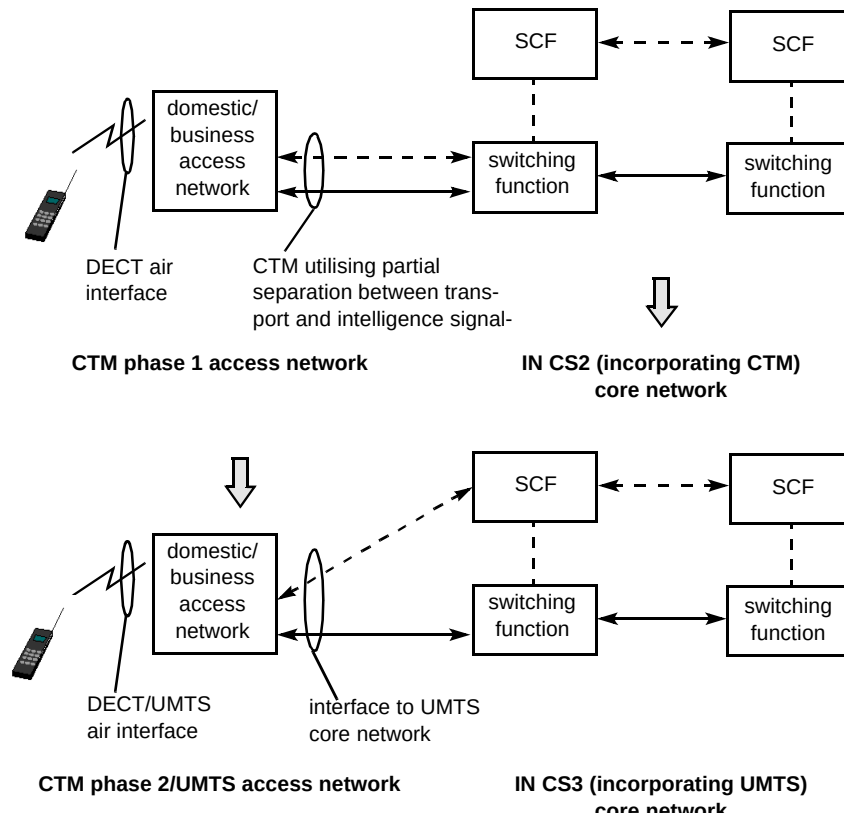


Fig 6 Fixed network migration to UMTS (including CTM).

progress. Ideally any subsequent developments of the CTM standards will be on the evolutionary path between the first phase of CTM and UMTS (second part of Fig 6).

to support mobility features, becoming the UMTS service and mobility layer. Supplementary services would continue to be offered to both networks using the evolved 'CAMEL' intelligence interfaces.

6.2 Evolution from GSM to UMTS

As described previously, UMTS is not intended to replace GSM, but to supplement the low data rates and fixed bandwidth capabilities supplied by GSM. Consequently, it can be envisaged that both systems may run in parallel for a considerable time, making direct evolution unnecessary. Additionally, complete evolution from GSM to a flexible bandwidth system such as UMTS is unlikely to be possible due to the inherent differences in the networks. Full evolution would require full protocol interworking to be supported at many nodes in both networks (the GSM BSS, MSC, etc) which would be uneconomical and would most likely result in a loss of functionality in both network parts. However, partial migration of the GSM network towards UMTS is possible and would allow closer integration of UMTS and GSM.

This partial migration towards UMTS, shown in Fig 7, concerns the support of customised supplementary services from a separate service layer. This is currently being standardised in GSM as the 'CAMEL' feature [4], providing IN capabilities in a mobile environment. With the introduction of UMTS, the service layer could be enhanced

Implementing this evolutionary development for GSM would produce an integrated UMTS/GSM system, reducing implementation costs for UMTS and strengthening service delivery capabilities in GSM. However, in order to achieve this goal, it is necessary that IN CS3 developments encompass the GSM 'CAMEL' feature in addition to providing UMTS specific developments.

7. Conclusions

UMTS technically aims to provide, through the appropriate development and convergence of UMTS, IN, GSM and B-ISDN standards, a system which will efficiently deliver:

- an integrated network solution which complements the existing 2nd generation standards (e.g. GSM within Europe),
- customer data rates up to 2 Mbit/s and flexible bandwidth allocation irrespective of whether the customer is fixed or mobile,

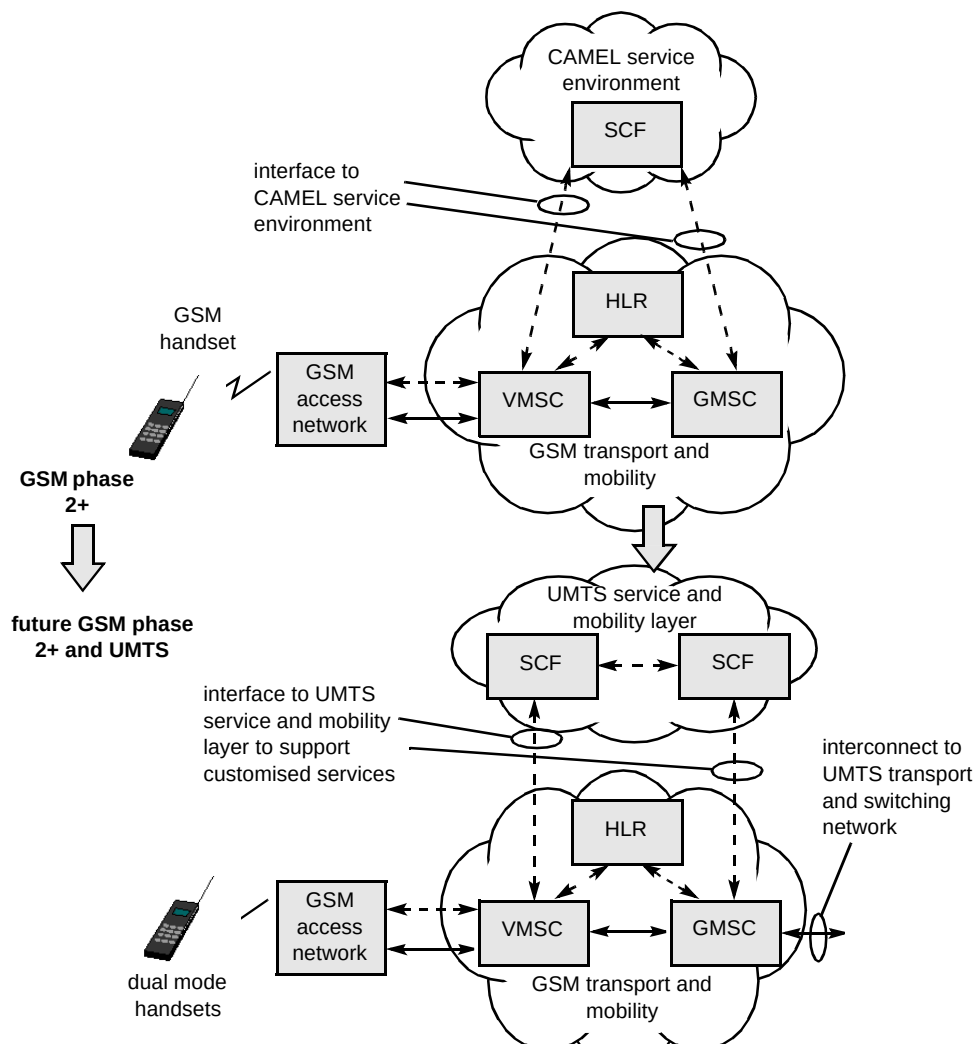


Fig 7 Migration of service support from GSM to UMTS.

- common supplementary services for customers roaming globally across a number of underlying transport networks (fixed, GSM, UMTS),
- an efficient network whereby network loading caused by mobility is removed from the switching and transport layer of the network,
- contiguous UMTS support in a number of access environments (any time, any place).

UMTS must also be a commercially viable system, designed to both integrate with and migrate from existing fixed and mobile systems. The level of integration or the degree of migration will ultimately be determined by the service requirements made upon UMTS. A cornerstone of these requirements is that all stakeholders (mobile operators, fixed operators and customers) desire an integrated service infrastructure giving operators and customers the ability to have the same core services whether

fixed or mobile, and irrespective of the environment. Obviously, there will always be differentiation between the service set offered over fixed access and mobile access because of market requirements (e.g. video on demand for mobile access would not appear viable), but the core services should be supported in similar ways on both networks. Some of the specific technical requirements to meet this goal have been discussed and, if these are realised, the cost of provisioning services and new applications for the next generation of mobile networks should be lowered for both network operators and customers, leading to further growth in the telecommunications industry as new applications become cost effective.

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architecture and its functionality over the medium term.



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(MoNet) project.